

## DANMARKSHAVN, Greenland

WMO: 043200

Lat: 76.769N

Lon: 18.668W

Elev: 12

StdP: 101.18

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 2             | -34.9      | -32.9 | -39.1                         | 0.1 | -33.8 | -37.0 | 0.1 | -31.8 | 25.9                  | -14.1 | 22.5 | -13.7 | 1.1                   | 300  | 1.083 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |     |      |     |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-----|------|-----|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%  |      | 2%  |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB  | MCWB | DB  | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e) | (f)  | (g) | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 7             | 5.0                    | 10.8            | 6.0  | 9.6 | 5.3  | 8.3 | 4.6  | 6.3                 | 10.0 | 5.6 | 9.0  | 4.8 | 7.8  | 4.9                  | 280  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 3.4                             | 4.8 | 6.1  | 2.7 | 4.6 | 5.5  | 2.2 | 4.4 | 5.1  | 21.2          | 10.3 | 19.7 | 9.2  | 18.2 | 7.9  | 8.6               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 19.1          | 14.5 | 11.9 | -37.7                      | 11.8 | 3.0                | 0.4 | -39.8                                              | 12.1 | -41.6      | 12.3 | -43.3      | 12.5 | -45.4      | 12.7 |     |     |
| (5)               |      |      | -37.8                      | 7.1  | 2.9                | 0.7 | -39.9                                              | 7.6  | -41.6      | 8.0  | -43.2      | 8.4  | -45.3      | 8.9  | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | -10.9       | -21.3 | -22.2 | -22.2 | -16.5 | -6.6  | 1.3   | 4.2   | 3.2   | -2.7  | -11.5 | -17.4 | -20.4 |  |  |
|                                                                        | DBStd       | 11.08       | 6.55  | 6.97  | 5.55  | 5.53  | 4.32  | 2.83  | 2.51  | 2.46  | 3.21  | 4.75  | 4.93  | 5.91  |  |  |
|                                                                        | HDD10.0     | 7643        | 969   | 903   | 999   | 794   | 516   | 259   | 180   | 212   | 380   | 667   | 823   | 942   |  |  |
|                                                                        | HDD18.3     | 10684       | 1227  | 1136  | 1257  | 1044  | 774   | 509   | 438   | 470   | 630   | 925   | 1073  | 1200  |  |  |
|                                                                        | CDD10.0     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDD18.3     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 4.2         | 5.0   | 4.7   | 4.6   | 3.5   | 3.4   | 3.2   | 3.2   | 3.3   | 4.1   | 5.0   | 5.3   | 5.3   |  |  |
|                                                                        | PrecAvg     | 163         | 18    | 18    | 18    | 13    | 6     | 5     | 15    | 17    | 13    | 14    | 17    | 17    |  |  |
| (15) Precipitation                                                     | PrecMax     | 381         | 93    | 74    | 59    | 47    | 36    | 24    | 50    | 64    | 45    | 55    | 57    | 62    |  |  |
|                                                                        | PrecMin     | 63          | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     |  |  |
|                                                                        | PrecStd     | 65          | 18    | 15    | 14    | 13    | 7     | 5     | 13    | 16    | 11    | 13    | 15    | 16    |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | -1.7  | -0.4  | -2.9  | 0.0   | 4.1   | 11.0  | 11.9  | 11.5  | 4.7   | 1.4   | -2.4  | -1.8  |  |  |
|                                                                        |             | MCWB        | -2.3  | -0.8  | -3.1  | -0.5  | 0.8   | 6.1   | 6.5   | 6.3   | 1.5   | 0.3   | -3.1  | -2.3  |  |  |
|                                                                        | 2%          | DB          | -5.1  | -3.4  | -8.5  | -3.2  | 1.6   | 9.2   | 11.2  | 10.1  | 3.3   | -1.0  | -5.5  | -5.3  |  |  |
|                                                                        |             | MCWB        | -5.6  | -3.8  | -9.0  | -4.1  | -0.5  | 5.0   | 6.3   | 5.6   | 1.0   | -2.2  | -6.3  | -6.2  |  |  |
|                                                                        | 5%          | DB          | -8.8  | -8.2  | -12.0 | -6.3  | 0.4   | 7.7   | 10.2  | 8.6   | 2.3   | -3.4  | -7.8  | -8.4  |  |  |
|                                                                        |             | MCWB        | -9.5  | -8.8  | -12.6 | -7.4  | -1.0  | 4.1   | 5.7   | 4.5   | 0.4   | -4.5  | -8.5  | -9.1  |  |  |
|                                                                        | 10%         | DB          | -11.8 | -12.2 | -14.4 | -8.7  | -0.6  | 6.0   | 9.0   | 7.2   | 1.4   | -5.1  | -9.9  | -11.8 |  |  |
|                                                                        |             | MCWB        | -12.4 | -12.8 | -15.0 | -9.7  | -1.9  | 3.2   | 5.1   | 3.9   | -0.3  | -6.1  | -10.6 | -12.6 |  |  |
|                                                                        | 0.4%        | WB          | -2.3  | -0.7  | -3.0  | -0.4  | 1.1   | 6.4   | 7.3   | 7.0   | 2.6   | 0.5   | -3.2  | -2.1  |  |  |
|                                                                        |             | MCDB        | -1.9  | -0.5  | -2.8  | 0.1   | 3.1   | 10.5  | 11.0  | 10.8  | 3.9   | 1.1   | -2.7  | -1.9  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB          | -5.6  | -3.7  | -9.1  | -4.1  | 0.1   | 5.2   | 6.6   | 5.9   | 1.6   | -2.2  | -6.3  | -6.1  |  |  |
|                                                                        |             | MCDB        | -5.2  | -3.2  | -8.6  | -3.2  | 1.3   | 8.8   | 10.4  | 9.3   | 2.6   | -1.2  | -5.5  | -5.4  |  |  |
|                                                                        | 5%          | WB          | -9.4  | -8.6  | -12.5 | -7.3  | -0.9  | 4.2   | 6.0   | 5.0   | 0.8   | -4.4  | -8.5  | -9.0  |  |  |
|                                                                        |             | MCDB        | -8.8  | -8.1  | -12.0 | -6.3  | 0.4   | 7.3   | 9.7   | 7.9   | 1.9   | -3.5  | -7.8  | -8.4  |  |  |
|                                                                        | 10%         | WB          | -12.4 | -12.6 | -15.0 | -9.6  | -1.9  | 3.2   | 5.2   | 4.2   | 0.0   | -6.1  | -10.6 | -12.5 |  |  |
|                                                                        |             | MCDB        | -11.9 | -12.1 | -14.4 | -8.7  | -0.7  | 5.8   | 8.5   | 6.6   | 1.3   | -5.1  | -9.9  | -11.8 |  |  |
|                                                                        | 5% DB       | MDBR        | 6.6   | 6.9   | 6.6   | 6.9   | 4.9   | 4.4   | 5.0   | 4.4   | 3.5   | 4.2   | 5.3   | 6.4   |  |  |
|                                                                        |             | MCDBR       | 8.3   | 8.7   | 8.2   | 7.5   | 4.8   | 7.0   | 7.4   | 6.5   | 4.0   | 5.0   | 6.6   | 8.0   |  |  |
|                                                                        |             | MCWBR       | 8.0   | 8.6   | 8.0   | 6.8   | 3.6   | 4.3   | 4.2   | 3.7   | 2.7   | 4.4   | 6.3   | 7.7   |  |  |
|                                                                        |             | MCWBR       | 8.0   | 8.6   | 8.2   | 7.0   | 4.5   | 6.7   | 7.1   | 6.0   | 3.7   | 4.6   | 6.4   | 7.9   |  |  |
| (35) Clear-Sky Solar Irradiance                                        | taub        |             | N/A   | 0.164 | 0.219 | 0.271 | 0.305 | 0.304 | 0.300 | 0.284 | 0.239 | 0.167 | N/A   | N/A   |  |  |
|                                                                        |             | taud        | N/A   | 1.854 | 2.089 | 2.084 | 2.051 | 2.163 | 2.327 | 2.401 | 2.295 | 1.867 | N/A   | N/A   |  |  |
|                                                                        | Edn at Noon |             | 0     | 134   | 660   | 789   | 819   | 844   | 836   | 793   | 658   | 93    | 0     | 0     |  |  |
|                                                                        |             | Edh at Noon | 0     | 9     | 53    | 93    | 117   | 110   | 89    | 69    | 45    | 7     | 0     | 0     |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.00  | 0.05  | 1.08  | 3.70  | 5.74  | 7.04  | 5.92  | 3.57  | 1.36  | 0.20  | 0.00  | 0.00  |  |  |
|                                                                        | RadStd      |             | 0.00  | 0.01  | 0.10  | 0.30  | 0.50  | 0.68  | 0.47  | 0.35  | 0.21  | 0.03  | 0.00  | 0.00  |  |  |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                        | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only           | +0.51 | N/A     | -1.20  | +1.09   | +0.68 | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page